Predicting Soil Organic Carbon Stock in Plateau Swamp Wetlands Using Multisource Remote Sensing and Spectral Measurements: A Case Study of the Dianchi Basin

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***Abstract*— Soil organic carbon density (SOCD) in swamp wetlands is a critical indicator for assessing global carbon stocks. In plateau wetlands, challenges such as dense vegetation cover and fragmented land distribution complicate SOCD research. The availability of high-resolution optical and radar satellite data introduces new possibilities for precise carbon stock predictions. This study proposes a framework that combines multisource remote sensing data with the sparrow search algorithm random forest (SSA-RF) algorithm to predict SOCD in plateau swamp wetlands. It also compares the effectiveness of laboratory spectroscopy and multisource remote sensing in monitoring SOCD. We integrated 24 features from Sentinel-1 (S1), Sentinel-2 (S2), topographic, and climatic data, along with spectral data ranging from 550 to 1400 nm, to construct the SSA-RF model and map the SOCD distribution of swamp wetlands in Dianchi Basin. Additionally, we estimated the total soil organic carbon (SOC) stock in these wetlands. The results indicate that the multisource remote sensing SSA-RF model (S1 + S2 + topographic + climatic SSA-RF) achieved an R^2 of 0.76, a root-mean-square error (RMSE) of 1.14, a mean absolute error (MAE) of 0.65, and a residual predictive deviation (RPD) of 1.98. Compared to the spectral model, this model improved the R^2 by 0.24 and the RPD by 0.5. Relative to the S1 + S2 SSA-RF model, the R^2 is increased by 0.15, and the RMSE is decreased by 0.46. The total SOC stock of the swamp wetlands in the Dianchi Basin was estimated to be 1.32×10^5 t. This study provides a new framework for predicting carbon stocks in plateau wetlands, offering a reference for global wetland carbon sink assessments.**

***Index Terms*— Multisource remote sensing, Sentinel-2 (S2), soil organic carbon density (SOCD), spectral information, swamp wetlands.**

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I. Introduction

WETLANDS, as one of the three major ecosystems on Earth, play a significant role in the global carbon cycle and carbon balance, forming a crucial part of the terrestrial carbon pool [1], [2]. Among the diverse types of wetlands, swamp wetlands are particularly noteworthy for their substantial carbon sequestration capabilities and significant carbon stocks [3]. The prevalent abundance of surface water in swamp wetlands creates an anoxic soil environment that persists for extended periods. This condition inhibits the complete decomposition of soil organic matter, thereby facilitating the sequestration of substantial amounts of organic carbon within swamp wetland soils [4]. Plateau swamp wetlands are a crucial component of the global wetland ecosystem, which is characterized by fragmentation and wide distribution [5], [6]. Studies indicate that over 90% of carbon stock in swamp wetlands are found in the soil, and even minor variations in soil carbon content can alter atmospheric carbon dioxide levels, thus potentially leading to global climate change [7], [8], [9]. Consequently, precise quantification of soil organic carbon density (SOCD) in plateau swamp wetlands is crucial for the sustainable management of wetlands and pivotal in maintaining the carbon balance within these ecosystems.

Traditional methods for monitoring SOCD typically involve field surveys, sampling, and laboratory measurements. While these methods are precise, they are also costly and time-consuming, limiting their scalability. Interpolation mapping methods based on spatial autocorrelation, such as kriging and inverse distance weighting, have also been used for extensive SOCD monitoring. However, due to the nonuniform distribution characteristics of wetlands, these methods struggle

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to ensure the reliability of predicted carbon stock [10], [11]. This article has utilized ecosystem service assessment models to estimate wetland carbon stock, such as the work by Deng et al. [12] and Zhang et al. [13], who employed the Integrated Valuation of Ecosystem Services and Tradeoffs (InVEST) model to estimate the carbon stock of wetlands in the Guangdong–Hong Kong–Macau Greater Bay Area (GBA) with notable success. Nonetheless, these approaches, which rely on fixed carbon densities for specific wetland types, may not yield accurate results. With the rapid development of computational power, machine learning algorithms offer a promising solution for quickly and accurately estimating SOCD on a large scale [14], [15]. These algorithms can effectively integrate diverse datasets to establish spatial prediction models for SOCD, offering accurate estimates of wetland soil organic carbon (SOC) storage. Popular SOCD prediction models include multiple linear regression, bagged classification and regression tree (CART), support vector machine (SVM), and random forest (RF) [16], [17]. However, these models are prone to issues such as susceptibility to local optima, sensitivity to parameter selection, and overfitting, making them less effective for SOCD monitoring in plateau regions [18], [19]. Therefore, this study utilizes the sparrow search algorithm (SSA), known for its excellent stability and superior global search capabilities, combined with the RF model, to develop an SSA-RF regression analysis model for plateau swamp SOCD prediction.

Spectral and remote sensing technologies are increasingly utilized for monitoring SOCD. Research indicates that spectral technology can effectively establish a direct relationship between soil spectral data and soil component content within specific wavelengths (350–2500 nm), facilitating rapid and precise estimations of SOCD [20], [21]. For instance, Wijewardane et al. [22] developed a comprehensive soil spectral library for the United States through field sampling, successfully predicting SOC content. Similarly, Gholizadeh et al. [23] demonstrated the superiority of proximal spectroscopy over drone and satellite spectroscopy in predicting SOCD in agricultural fields. Remote sensing technology enables large-scale, synchronous soil observations, which are crucial for extensive predictions of SOCD. In recent years, data from various optical and radar imaging sources have been extensively utilized to assess surface soil characteristics and facilitate regional-scale SOCD predictions [23], [24], [25], [26]. However, previous studies have predominantly focused on agricultural areas with uniform plot distributions and coastal mudflat wetlands characterized by sparse vegetation. In contrast, plateau swamp wetlands exhibit greater spatial heterogeneity in SOCD distribution due to their complex topography and fragmented plots, which contribute to significant spatial heterogeneity in SOCD distribution. Moreover, the presence of abundant surface vegetation further complicates the accuracy of these predictions. Given these challenges, it is imperative to explore the suitability and effectiveness of proximal spectral and multisource remote sensing technologies specifically tailored for predicting SOCD in the unique environments of plateau swamp wetlands.

Selecting appropriate spectral bands is crucial for the proximal spectral prediction of SOCD. To enhance the reflection and absorption characteristics of spectral curves, researchers often apply mathematical transformations to raw spectral reflectance data, such as first derivative transformations, reciprocal transformations, and logarithmic transformations. These methods help to identify spectral curves that are highly correlated with SOCD, thereby effectively harnessing spectral information for predictive SOCD [27], [28], [29]. In parallel, efforts to predict SOCD using multisource data have focused on identifying the optimal feature input variables. For example, research by Zhang et al. [30] and Wang et al. [31] utilizing Sentinel-1 (S1) imagery to predict regional SOC stock has found S1 data to be an exceptionally effective resource for SOCD prediction. Similarly, Zhou et al. [32] have combined S1 and Sentinel-2 (S2) data with a digital elevation model (DEM) to predict SOCD, revealing terrain variables as significant explanatory variables for SOCD predictions. Their findings suggest that the inclusion of DEM data enriches the feature set, substantially improving the accuracy of predictions. However, the potential influence of high-precision climate indicators on the prediction effectiveness of SOCD in plateau wetlands remains underexplored. Expanding research to include these indicators could provide a more comprehensive understanding and improve the predictive capabilities of SOCD in these complex environments.

This study aims to utilize small-sample SOCD field data along with the SSA-RF algorithm to achieve the following objectives: 1) compare the predictive effectiveness of laboratory spectral measurement techniques and multisource remote sensing technologies in estimating SOCD in plateau swamp wetlands; 2) explore the most suitable prediction framework for SOCD in these wetlands; and 3) map the SOCD distribution and estimate carbon stock across the Dianchi Basin.

II. Study Area and Data

A. Study Area

The Dianch Basin, centrally positioned on the Yunnan–Guizhou Plateau in southwestern China, spans 102°29′E–103°1′E and 24°29′N–25°28′N (see Fig. 1). It is a plateau lake basin centered around Dianchi Lake. The Yangtze River, Pearl River, and Red River flow through this basin. The terrain gradually slopes from North to South, covering an area of 2920 km^2, with elevations ranging from a minimum of 1803 m to a maximum of 2781 m. The region features a topographic corridor shaped by a narrow lake basin and enclosed by mountains. This area is frequently shrouded in clouds and experiences a low-latitude subtropical highland monsoon climate. The average annual temperature is 15 °C with precipitation totaling approximately 1001.6 mm. The climate conditions provide abundant sunshine, minimal frost, and a prolonged vegetation growth period [33]. According to data from China's Second Soil Survey, the soils in the study area are primarily lateritic and yellow soils. The basin contains numerous narrow rivers and vegetation-covered swamps. In recent years, the Chinese

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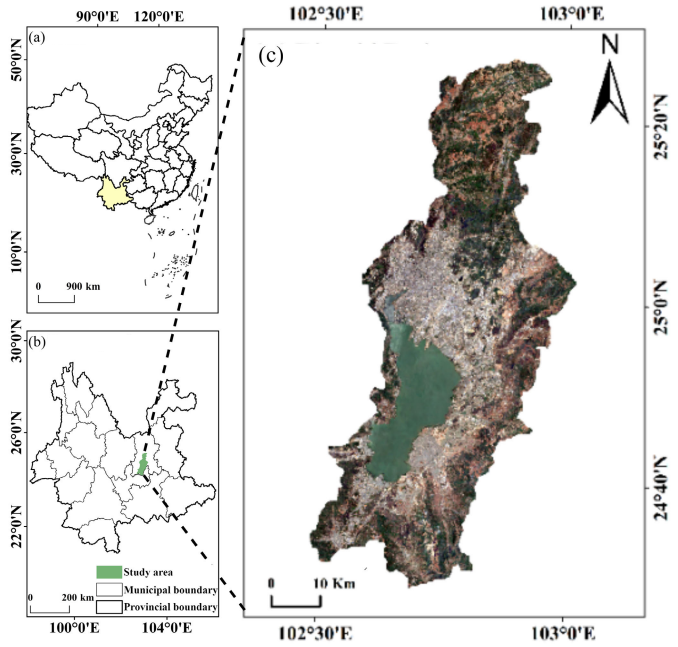


Fig. 1. Location of the Dianchi Basin. (a) Location of Yunnan Province in China. (b) Location of the study area. (c) S2 image of the study area (R: Band 4, G: Band 3, and B: Band 2).

government has effectively protected the wetland resources in the basin, supporting the establishment of more than ten wetland parks, including the Baofeng National Wetland Park and the Xiliangtang National Wetland Park. The abundance of wetland resources in the study area provided favorable conditions for our experiments.

B. Datasets

In this study, we employed remote sensing imagery, including Sentinel-1A and Sentinel-2A. The Sentinel-1A images, in interferometric wide swath (IW) mode and dual-polarization (VV and VH) ground range detected (GRD) format, were acquired from the European Space Agency (ESA) on November 1 and 13, 2023. Concurrently, two scenes of Sentinel-2A multispectral instrument (MSI) level-2A (L2A) images covering the study area on November 2 and 15, 2023, were downloaded, with cloud cover less than 10% and closely matched to the dates of soil sample collection. Wetland maps for this article were sourced from the CAS_wetlands datasets [34], employing a hybrid classification method that combines object-based and hierarchical approaches on 30-m resolution Landsat images, ensuring an overall accuracy of 95.1%. We refined the 1-m resolution wetland distribution data for the Dianchi Basin from Cai et al. [33], to delineate the final wetland distribution data in our study area. Topographic data were obtained from NASA's latest global 30-m-resolution DEM data (NASA DEM). Climate datasets, including 2020 temperature and precipitation data at a 30-m resolution, were acquired from the Chengdu Institute of Mountain Hazards and Environment's High-Resolution Mountain Environment Mapping Project (FRMM). These datasets were downscaled from ERA5_LAND data to 1-km resolution using the delta temperature data and were adjusted for vertical lapse rate and slope aspect solar radiation effects to minimize elevation and terrain influences. The annual precipitation dataset showed an average error of 34.3 mm (0.094 mm/day) and a mean relative error of 6.9%.

III. Methodology

This study employs the SSA-RF algorithm to establish a prediction framework for SOCD in plateau swamp wetlands (see Fig. 2), utilizing both measured spectral and multisource remote sensing data. We compare the effectiveness of laboratory spectral measurements with that of multisource remote sensing in predicting SOCD. Additionally, this article undertakes mapping the SOCD distribution and calculates the carbon stock within the Dianchi Basin.

A. Soil Sample Collection and Measurement

Based on a wetland distribution dataset and field surveys, we selected 103 representative soil sampling sites, as depicted in Fig. 3(a). Throughout November 2023, we collected surface soil samples (0–20 cm) from these locations. The sampling process involved the use of a soil auger, with three soil samples uniformly collected and mixed at each site to ensure an accurate representation of soil characteristics. Additionally, a ring knife was used to specifically collect samples for determining soil bulk density. The geographical coordinates of each sampling site were precisely recorded using a handheld global positioning system (GPS). Upon returning to the laboratory, the freshly collected soil samples were air-dried. Then, we measured the soil bulk density and gravel content, removing all nonsoil materials such as plant roots and rocks. The samples were ground and successively sieved through 0.25- and 0.098-mm meshes. We used an enviro total organic carbon (TOC) analyzer to perform four parallel measurements on each soil sample to determine its organic carbon content. The average of these measurements was then used to establish the SOCD. The calculation of SOCD was conducted as follows [35]:

$$\mathrm{SOCD} = c \times d \times r \times (1\text{-D}) \tag{1}$$

where SOCD represents the SOC density (kg·C/m^2), c denotes the percentage of organic carbon content in the soil, d refers to the soil layer thickness (m), r indicates the soil bulk density (kg/m^3), and D is the proportion of gravel with a diameter greater than 2 mm.

B. Soil Spectral Reflectance Measurement

We measured the spectral reflectance of soil samples using the Avantes AvaField-3 portable spectroradiometer, which spans a wavelength range from 300 to 2500 nm. This instrument provides a spectral resolution of 1.5 nm from 300 to 1100 nm and 15 nm from 1100 to 2500 nm, with sampling intervals of 0.6 and 6 nm in these respective ranges. Prior to measurements, the spectroradiometer and its lamp were preheated for 30 min to ensure stability. For accurate reflectance measurement, soil samples were evenly spread in a 9-cm-diameter Petri dish to a depth of 2 cm. The surface of the samples was leveled with a stainless-steel blade to align flush with the Petri dish's rim,

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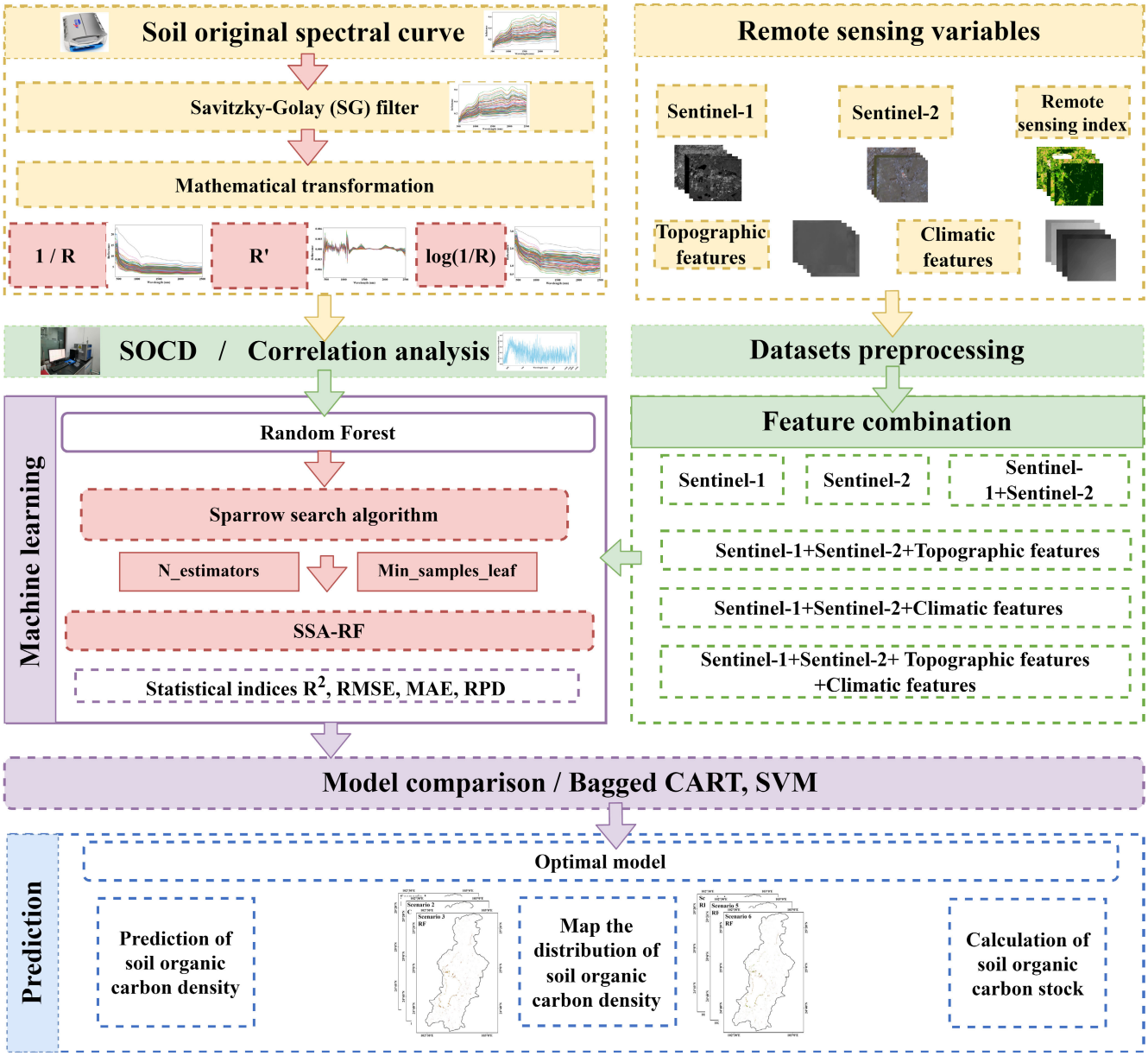


Fig. 2. Research framework.

optimizing the soil surface for maximum reflectivity and improved signal-to-noise ratio. Measurements were performed in a dark room to minimize stray light interference, with ten repeated measurements taken for each sample. The results were then averaged to produce the final spectral curve for each sample. The spectroradiometer was calibrated with a standard whiteboard before the initial scan and after every six measurements to maintain measurement accuracy [23], [36].

C. Dataset Preprocessing

1) Spectral Data Preprocessing: During the collection of spectral data, the raw spectral readings are often affected by changes in the external environment and the variable energy response of the spectrometer across different wavelengths. This variability introduces noise into the data. To ensure the accuracy of subsequent analyses and the robustness of model development and validation, we have applied preprocessing and mathematical transformations to the raw spectral data. Initially, we employed a Savitzky–Golay (SG) filter to smooth the raw data, aiming to eliminate high-frequency random noise [37]. After denoising, further manipulations such as reciprocal transformation ($1/R$), derivative transformation (R'), and reciprocal logarithmic transformation [$\log(1/R)$] were performed. These steps are designed to reduce measurement errors and enhance the extraction of spectral information, thereby improving the quality and utility of the spectral data for environmental analysis.

2) Remote Sensing Data Preprocessing: In this study, we utilized SNAP 8.0 to conduct orbital calibration, radiometric calibration, speckle filtering, and terrain correction on S1 data. Following these preprocessing steps, we transformed the amplitude information from the synthetic aperture radar

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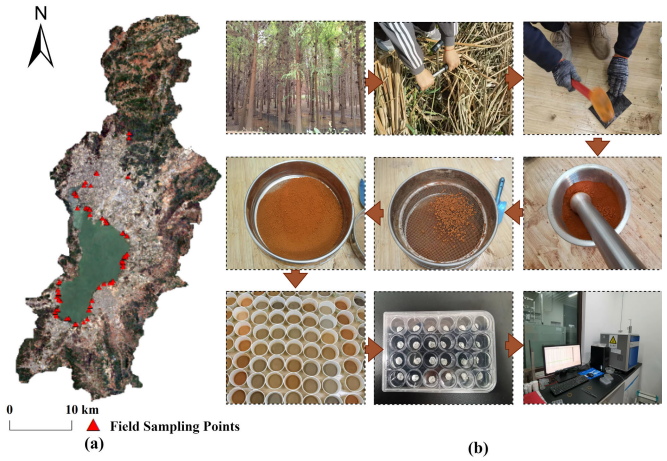


Fig. 3. Data collection of field samples. (a) Location of field sample collection. (b) Organic carbon content measurement process.

images into backscatter coefficients. Then, we extracted the backscatter coefficients for VV and VH polarizations to serve as predictive features. Additionally, we utilized ENVI 5.3 to perform radiometric and atmospheric corrections on two S2 images [38]. Following corrections, the images were mosaicked and cropped to fully cover the study area. From these S2 images, ten spectral bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, and B12) and seven spectral indices closely related to SOCD were extracted as predictive factors (see Table I) [23]. To explore the impact of different environmental characteristics on the prediction of SOCD, we designed six feature combination scenarios (see Table II): S1 (Scenario 1), S2 (Scenario 2), S1 + S2 (Scenario 3), S1 + S2 + topographic (Scenario 4), S1 + S2 + climatic (Scenario 5), and S1 + S2 + topographic + climatic (Scenario 6).

D. Model Prediction

We developed a model using the SSA-RF algorithm, based on data from 103 field-measured samples. The SSA is an optimization technique modeled after the foraging patterns of sparrows [39]. For the 103 field soil measurement points obtained, we selected 80% for model training and 20% for model testing.

1) Sparrow Search Algorithm: In the SSA algorithm, the sparrow population is categorized into three roles: discoverers, joiners, and scouts. The position update formula for the discoverers is as follows [39], [40]:

$$X_{i.j}^{t+1} = \begin{cases} X_{i.j}^{t} \cdot \exp\left(\dfrac{-i}{\alpha \cdot M}\right), & F < S \\ X_{i.j}^{t} + Q \cdot L, & F \geq S \end{cases} \tag{2}$$

where $X_{i.j}^{t}$ represents the position of the ith sparrow in the jth dimension during the tth iteration, with $j \in [1, d]$; α is a random number within the range [0, 1]; M denotes the maximum number of iterations; Q is a random number drawn from a standard normal distribution; F and S represent the alarm and safety values, respectively; and L is a $1 \times d$ matrix.

The position update formula for the joiners is as follows:

$$X_{i.j}^{t+1} = \begin{cases} Q \cdot \exp\left(\dfrac{X_{w}^{t} - X_{i.j}^{t}}{i^2}\right), & i > \dfrac{n}{2} \\ X_{P}^{t+1} + |X_{i.j}^{t} - X_{P}^{t+1}| \cdot A^{+} \cdot L, & i \leq \dfrac{n}{2} \end{cases} \tag{3}$$

where X_P and X_W denote the optimal and worst positions, respectively, and A is a matrix of $1 \times d$ with each element randomly assigned a value of either 1 or -1. Additionally, $A^{+} = A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}$. The position update formula for the scouts is as follows:

$$X_{i.j}^{t+1} = \begin{cases} X_{b}^{t} + \beta \cdot |X_{i.j}^{t} - X_{b}^{t}|, & f_i > f_g \\ X_{i.j}^{t} + K \cdot \left(\dfrac{|X_{i.j}^{t} - X_{w}^{t}|}{(f_i - f_w) + \varepsilon}\right), & f_i = f_g \end{cases} \tag{4}$$

where X_b is the global optimal position; K (where $K \in [-1, 1]$) is a random number that indicates the direction of sparrow movement; β is a random number drawn from a normal distribution with mean 0 and variance 1, used to control the step size; f_i is the current fitness value of individual sparrow; and f_g and f_w are the current fitness values of the global best and worst individuals, respectively. ε is a constant and set to a nonzero value.

2) SSA-RF Algorithm: RF is an ensemble learning algorithm composed of multiple decision trees, which enhances predictive capability through bootstrap sampling with replacement and random feature selection [41]. In this study, we employ the SSA to globally optimize the n_estimators and min_samples_leaf parameters of the RF model. The n_estimators parameter determines the number of trees in the forest, while the min_samples_leaf parameter specifies the minimum number of samples required at each leaf node. Both parameters significantly affect the model's performance and its susceptibility to overfitting. The specific optimization steps are as follows: 1) population initialization; 2) obtain initial fitness values and identify the global optimum fitness value to determine the globally optimal population; 3) update the positions of discoverers, joiners, and scouts within the sparrow population; 4) acquire fitness values and the optimal population, and update sparrow positions; 5) check if the termination criteria are met; if not, repeat steps 2–4; and 6) train the RF model using the optimized parameters n_estimators and min_samples_leaf.

E. Model Evaluation

For the assessment of our predictive models for SOCD in plateau swamp wetlands, we constructed separate RF models using proximal spectroscopy techniques and multisource remote sensing data, with the parameters optimized by the SSA algorithm. The performance of these models was evaluated using tenfold cross-validation, and four validation metrics were established: R^2 (coefficient of determination), root-mean-square error (RMSE), mean absolute error (MAE), and residual predictive deviation (RPD). The methodologies for calculating these metrics are outlined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(O_i - P_i)^2}{\sum_{i=1}^{n}\left(O_i - \overline{O}\right)^2} \tag{5}$$

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TABLE I
ALL FEATURES USED IN THIS ARTICLE

	Feature Name	Description
Image features	Sentinel-1	VV, VH
	Sentinel-2	B2-B8, B8A, B11-12
Vegetation Index	NDVI	(NIR – RED) / (NIR + RED)
	RVI	NIR / RED
	EVI	2.5×(NIR-RED) / (NIR+6×RED-7.5×BLUE+1)
Water Index	NDWI	(GREEN-NIR) / (GREEN+NIR)
	MNDWI	(GREEN - SWIR) / (GREEN + SWIR)
Soil Index	SAVI	(NIR - RED) / (NIR + RED + L) × (1 + L)
	LSWI	(NIR - SWIR) / (NIR + SWIR)
Topographic features	DEM	SRTM 30
	Slope	Generated from DEM
	TWI	Generated from DEM
Climatic features	Precipitation (Precip)	FRMM
	Temperature (Temp)	FRMM

TABLE II
FEATURE COMBINATION SCENARIOS

Scenario	Description
Scenario 1	Sentinel-1
Scenario 2	Sentinel-2
Scenario 3	Sentinel-1+Sentinel-2
Scenario 4	Sentinel-1+Sentinel-2+topographic
Scenario 5	Sentinel-1+Sentinel-2+climatic
Scenario 6	Sentinel-1+Sentinel-2+topographic+climatic

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(O_i - P_i)^2} \tag{6}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|P_i - O_i| \tag{7}$$

$$\mathrm{RPD} = \frac{\mathrm{SD}}{\mathrm{RMSE}} \tag{8}$$

where n is the number of soil samples; P_i, O_i, and $\overline{O}$ represent the predicted value, observed value, and mean of the observed value, respectively; and SD represents the standard deviation of the measured value.

F. Model Comparison

1) SVM Regression Model: SVM is a sophisticated supervised learning algorithm extensively used for both classification and regression tasks. It addresses nonlinear data challenges by utilizing kernel functions. The principle of SVM involves mapping data to a higher dimensional space and identifying an optimal hyperplane that segregates the data into distinct categories while maximizing the margin between these categories.

2) Bagged CART Regression Model: CART is a flexible, nonparametric data mining technique that has been widely applied in domains such as soil property prediction. Research by Guo et al. [42] and Heung et al. [43] has demonstrated the high accuracy of CART in digital soil mapping. CART recursively partitions data using a binary splitting algorithm, thoroughly exploring the complex relationships between response variables and predictors. To enhance predictive performance and reduce the risk of overfitting, Bagged CART combines the original CART with bagging techniques, a strategy proven effective in multiple studies.

G. SOCD Distribution and Carbon Stock Calculation

We employed the optimized remote sensing model to map the distribution of SOCD in the swamp wetlands of the Dianchi Basin. Concurrently, we calculated the total carbon stock of the wetlands within this basin. The formula used for calculating the carbon stock is defined as follows [44]:

$$C_i = \mathrm{SOCD}_i \times A_i \tag{9}$$

$$C_{\mathrm{total}} = \sum_{i=1}^{n} C_i \tag{10}$$

where SOCD_i represents the carbon density of the ith pixel, A_i denotes the area of the ith pixel, C_i is the carbon stock of the ith pixel, n is the total number of pixels in the study area, and C_{total} is the total carbon stock.

IV. RESULTS

A. Descriptions of Soil Characteristics

After conducting a statistical analysis of SOCD in the study area, we have summarized the descriptive statistical results in Table III. These results demonstrate significant variability in the SOCD of soil samples, with a minimum value of 0.233 kg·C/m^2 and a maximum of 16.105 kg·C/m^2, highlighting the broad differences in soil properties across the study area. The mean SOCD is 2.468 kg·C/m^2, and the median is 1.979 kg·C/m^2, the latter providing a more stable indicator of central tendency for SOCD due to its insensitivity to extreme values. The standard deviation is 2.331 kg·C/m^2, reflecting the dispersion of sample values around the mean and revealing high volatility in SOCD. The coefficient of

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TABLE III
SOIL CHARACTERISTICS

	Min (kg C/m^2)	Me (kg C/m^2)	Max (kg C/m^2)	Med (kg C/m^2)	SD (kg C/m^2)	SE (kg C/m^2)	V	CV (%)
SOCD	0.233	2.468	16.105	1.979	2.331	0.230	5.434	93.982

Min= Minimum; Me= Mean; Max= Maximum; Med= Median; SD=Standard deviation; V= Variance; SE= Standard error; CV= Coefficient of variation.

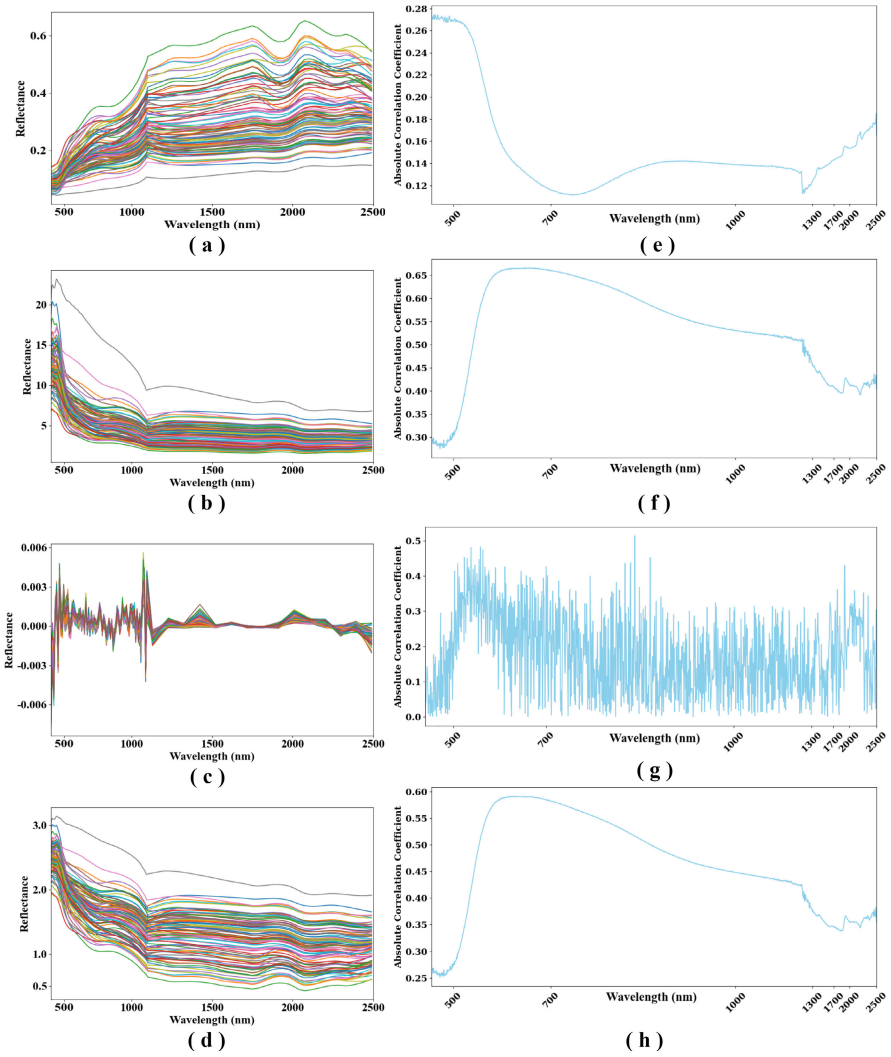


Fig. 4. (a)–(d) Filtered spectral and mathematical transformation curves (each curve represents a different sample), with (e)–(h) corresponding SOCD absolute correlation curves.

variation reaches 93.982%, further confirming the significant variability in SOCD among different samples. Additionally, the standard error is 0.23%, offering valuable information about the accuracy of the average SOCD estimate.

B. Soil Spectral Curves and Characteristic Wavelengths

We initially applied SG filtering to the raw spectral curves, aiming to smooth the data and reduce noise, as shown in Fig. 4(a). Subsequently, we performed a $1/R$ transformation [Fig. 4(b)], R' transformation [Fig. 4(c)], and $\log(1/R)$ transformation [Fig. 4(d)] on the filtered data to explore the effects of these mathematical transformations on enhancing the correlation between spectral data and SOCD. Fig. 4(e)–(h), respectively, represents the curves of the absolute values of the correlations between SOCD and the results of SG filtering, $1/R$, R', and $\log(1/R)$. Our analysis revealed that the $1/R$ transformation significantly improved the absolute correlation with SOCD, achieving a maximum coefficient of 0.65 [see Fig. 4(f)]. Based on this finding, we selected spectral curves with an absolute correlation greater than 0.45 (550–1400-nm wavelength band) after the $1/R$ transformation for predictive modeling of SOCD.

C. Model Comparison

1) Spectral Model: When comparing the performance of three algorithms, SSA-RF, CART, and SVM, in predicting

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TABLE IV
COMPARISON OF DIFFERENT MODELS FOR PREDICTING SOCD BASED ON SPECTRAL DATA

	R^2	RMSE	MAE	RPD
SSA-RF	0.52	2.34	1.42	1.48
SVM	0.51	2.87	1.49	1.21
CART	0.39	2.73	1.57	1.27

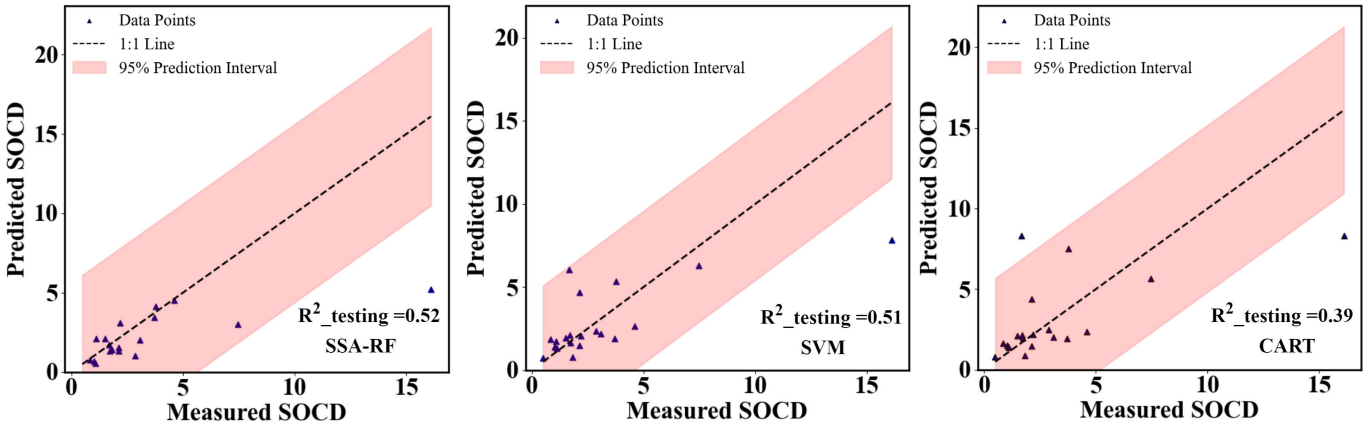


Fig. 5. Prediction of SOCD scatterplot based on spectral data.

TABLE V
COMPARISON OF DIFFERENT SCENARIOS BASED ON MULTISOURCE REMOTE SENSING FOR PREDICTING SOCD

		R^2	RMSE	MAE	RPD
Scenario1	(SSA-RF)	0.47	1.96	1.04	1.32
	(SVM)	0.29	4.11	3.33	1.19
	(CART)	0.46	1.90	0.98	0.68
Scenario2	(SSA-RF)	0.55	1.45	0.71	1.47
	(SVM)	0.59	2.10	6.68	1.24
	(CART)	0.60	1.61	0.69	0.78
Scenario3	(SSA-RF)	0.61	1.60	0.85	1.59
	(SVM)	0.24	4.22	3.42	1.31
	(CART)	0.59	1.86	0.70	0.66
Scenario4	(SSA-RF)	0.64	1.23	0.60	1.60
	(SVM)	0.64	1.44	2.25	1.19
	(CART)	0.61	1.55	0.63	0.78
Scenario5	(SSA-RF)	0.70	1.50	0.66	1.69
	(SVM)	0.70	1.72	3.96	1.18
	(CART)	0.69	1.59	0.56	0.83
Scenario6	(SSA-RF)	0.76	1.14	0.65	1.98
	(SVM)	0.73	1.60	3.12	1.17
	(CART)	0.71	1.19	0.69	0.84

SOCD based on spectral data, Table IV shows that the SSA-RF model outperforms the others. Fig. 5 displays scatterplots of the test dataset and the predicted dataset for the three prediction models. Specifically, the SSA-RF model has an R^2 value of 0.52, which means it can explain 52% of the total variance, reflecting a strong fit to the data. Additionally, its RPD value of 1.48 indicates reasonable predictive capability. The model also reports relatively low RMSE and MAE values, at 2.34 and 1.42, respectively, suggesting minor deviations between the predicted and actual values. In contrast, the SVM and CART models are less effective, with the CART model showing a particularly low R^2 of only 0.39, significantly lower than both the SSA-RF and SVM models. According to general performance evaluation criteria, models with an R^2 greater than 0.5, an RPD greater than 1.4, and lower values of RMSE and MAE are considered satisfactory. Thus, the SSA-RF model demonstrates superior performance in predicting SOCD content using spectral data and is the preferred choice among the three algorithms.

2) Multisource Remote Sensing Model: We explored the efficacy of various combinations of remote sensing data for predicting SOCD. The specific scenarios investigated include Scenario 1–Scenario 6, and the prediction models employed were SSA-RF, SVM, and CART (see Table V). Additionally, we plotted the variation of metric indicators as the feature combinations changed, as shown in Fig. 6. For Scenario 1,

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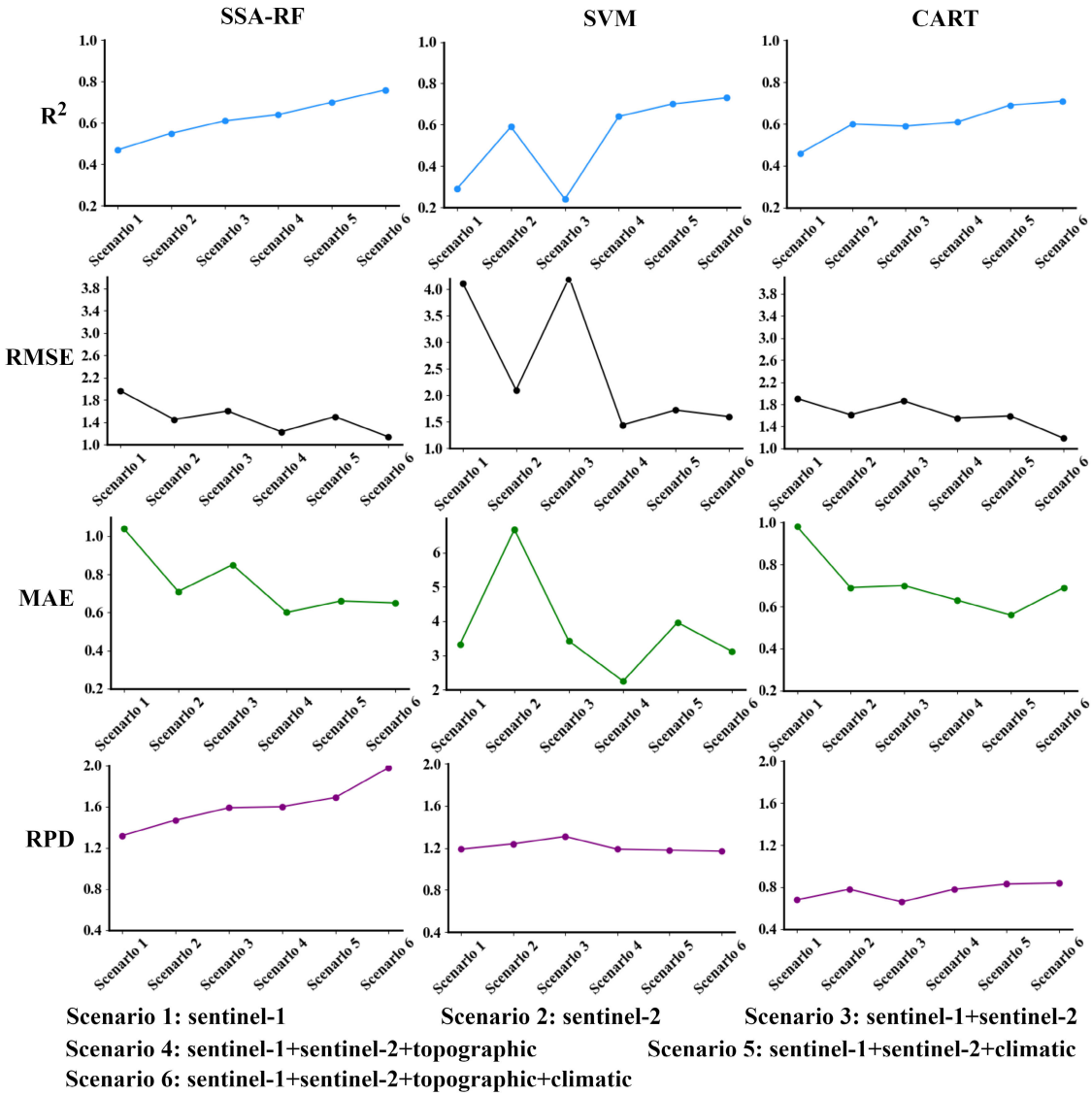


Fig. 6. Variation of metric indicators as the feature combinations changed.

we found that the SSA-RF model exhibited a relatively low coefficient of determination ($R^2 = 0.47$), with RMSE at 1.96, MAE at 1.04, and RPD at 1.32. This may be due to the limited number of radar features, which are insufficient to fully capture the characteristics related to SOCD. In Scenario 2, the CART model performed best, showing an R^2 of 0.60, RMSE of 1.61, MAE of 0.69, and RPD of 0.78, indicating that the spectral and index features based on S2 positively impact SOCD prediction. After combining S1 and S2 features (Scenario 3), the predictive performance of the SSA-RF model for SOCD improved ($R^2 = 0.61$), but there was a slight decline in the performance of CART and SVM models, likely due to model uncertainty. The introduction of topographic and climatic features (Scenarios 4 and 5) improved the predictive accuracy of all models (SSA-RF, SVM, and CART). The best-performing SSA-RF model, for instance, saw an increase in R^2 from 0.61 in Scenario 3 to 0.64 in Scenario 4 and 0.70 in Scenario 5, reflecting an enhancement in prediction accuracy. RPD also increased from 1.59 to 1.60 in Scenario 4 and to 1.69 in Scenario 5, showing that model stability was improving. The inclusion of climatic features led to a greater enhancement in model performance compared to the addition of topographic features. In Scenario 6, the predictive performance of the SSA-RF, SVM, and CART models was optimal, with the SSA-RF model exhibiting the best performance: $R^2 = 0.76$, RMSE = 1.14, MAE = 0.65, and RPD = 1.98. This demonstrates that the model's ability to explain SOCD variability significantly increased after integrating both topographic and climatic features. Compared to the spectral model, the multisource remote sensing model provided a stronger capability for predicting SOCD, with the SSA-RF model in Scenario 6 showing an increase from an R^2 of 0.52–0.76 and an RPD from 1.48 to 1.98, while RMSE and MAE decreased by 1.2 and 0.77, respectively. These results

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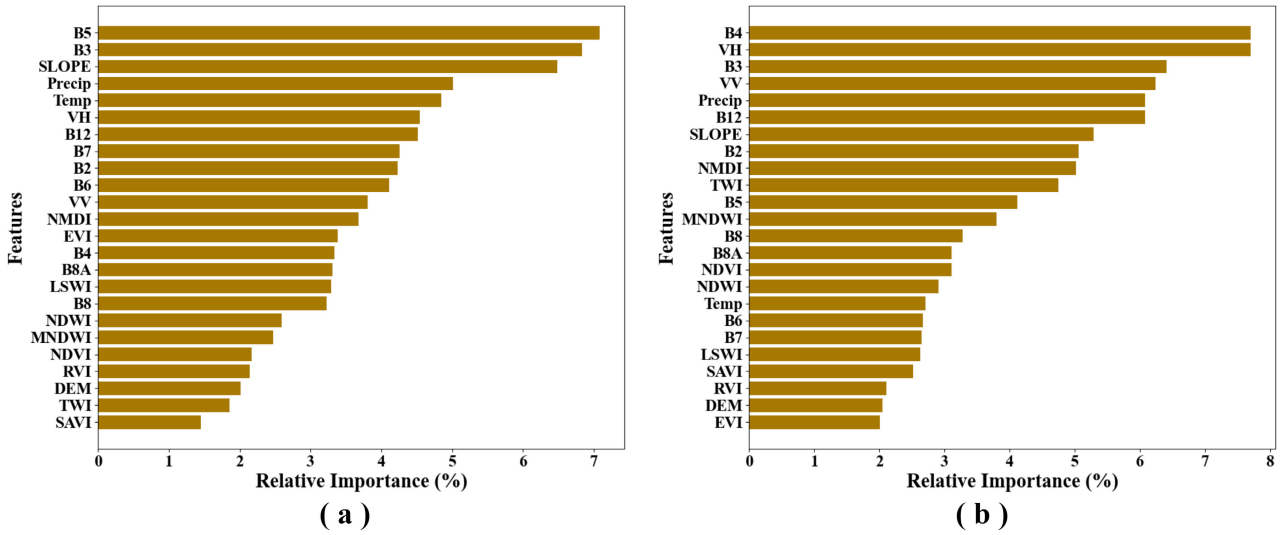


Fig. 7. Importance of the features. (a) SSA-RF. (b) CART.

consistently demonstrate that in plateau swamp areas with dense vegetation, the S1 + S2 + topographic + climatic SSA-RF algorithm significantly enhances both the accuracy and reliability of SOCD predictions.

D. Variable Importance in Multisource Remote Sensing Model

A detailed analysis of feature importance (percentage importance) for the SSA-RF and CART models is depicted in Fig. 7. The analysis revealed overall similarities between the SSA-RF and CART models in evaluating different environmental variables, yet with some differences. These findings underscore the significant contribution of environmental features to the prediction of SOCD. In the SSA-RF model, S2 and its derived features accounted for 62.09% of the total importance, while S1 features contributed 8.35%. Topographic and climatic features were responsible for 10.36% and 9.86% of the importance, respectively, with B5, B3, SLOPE, Precipitation, and Temperature emerging as the five most critical features. This underscores the dominant role of optical remote sensing data in SOCD prediction and highlights the significance of topographic and climatic features. For the CART model, the importance of S2 and its derived features was 65.19%, with S1 feature importance increasing to 13.94%. The contributions from topographic and climatic features were 12.09% and 8.79%, respectively. The top five influential features in the CART model were B4, VH, B3, VV, and Precipitation. These findings affirm the effectiveness of S1 and S2 data in predicting SOCD and further elucidate the valuable contributions of topographic and climatic information.

E. SOCD Spatial Distribution and Carbon Stock

We selected the optimal model from six different scenarios based on multisource remote sensing to analyze and research the spatial distribution of SOCD and the impact of various environmental features on this distribution. Fig. 8 illustrates the SOCD distribution across the Dianchi Basin swamp wetlands under different scenarios, while Fig. 9 provides detailed views of these distributions. Notably, the SSA-RF model in Scenario 6 (S1 + S2 + topographic + climatic SSA-RF) displays higher continuity and a concentration of high SOCD values, particularly near Dianchi Lake. This suggests that the SSA-RF model, when combined with topographic and climatic features, more effectively captures the complexities and subtle variations in SOCD distribution in plateau lake basin wetlands. Compared to other scenarios, Scenarios 1 and 3 show more dispersed high SOCD values, indicating that the feature sets employed did not fully capture all critical factors affecting SOCD. Scenario 2 utilized the CART model, which appears to have a more dispersed spatial distribution than the SSA-RF, suggesting less continuity in SOCD. Scenarios 4 and 5 also employed the SSA-RF model, and their predictions, compared to Scenario 6, show fewer high SOCD areas. This suggests that the predictive ability of remote sensing data combined with either single climatic or topographic data varies for SOCD in plateau swamp wetlands. Ultimately, the SSA-RF model from Scenario 6 was selected for its superior predictive performance. Using (9) and (10), we calculated the SOC stock of the study area, determining that the total carbon stock of the soil in the Dianchi Basin swamp wetlands amounts to 1.32×10^5 t.

V. Discussion

A. Performance of SOCD Prediction Models

This study's predictive indicators demonstrate that the spectral model is feasible for SOCD prediction in plateau wetlands, achieving an R^2 value above 0.5. However, the performance of predictions using multisource remote sensing data significantly surpasses those reliant solely on laboratory spectral model. The choice of different remote sensing data, combinations of environmental variables, and machine learning algorithms all influence model performance.

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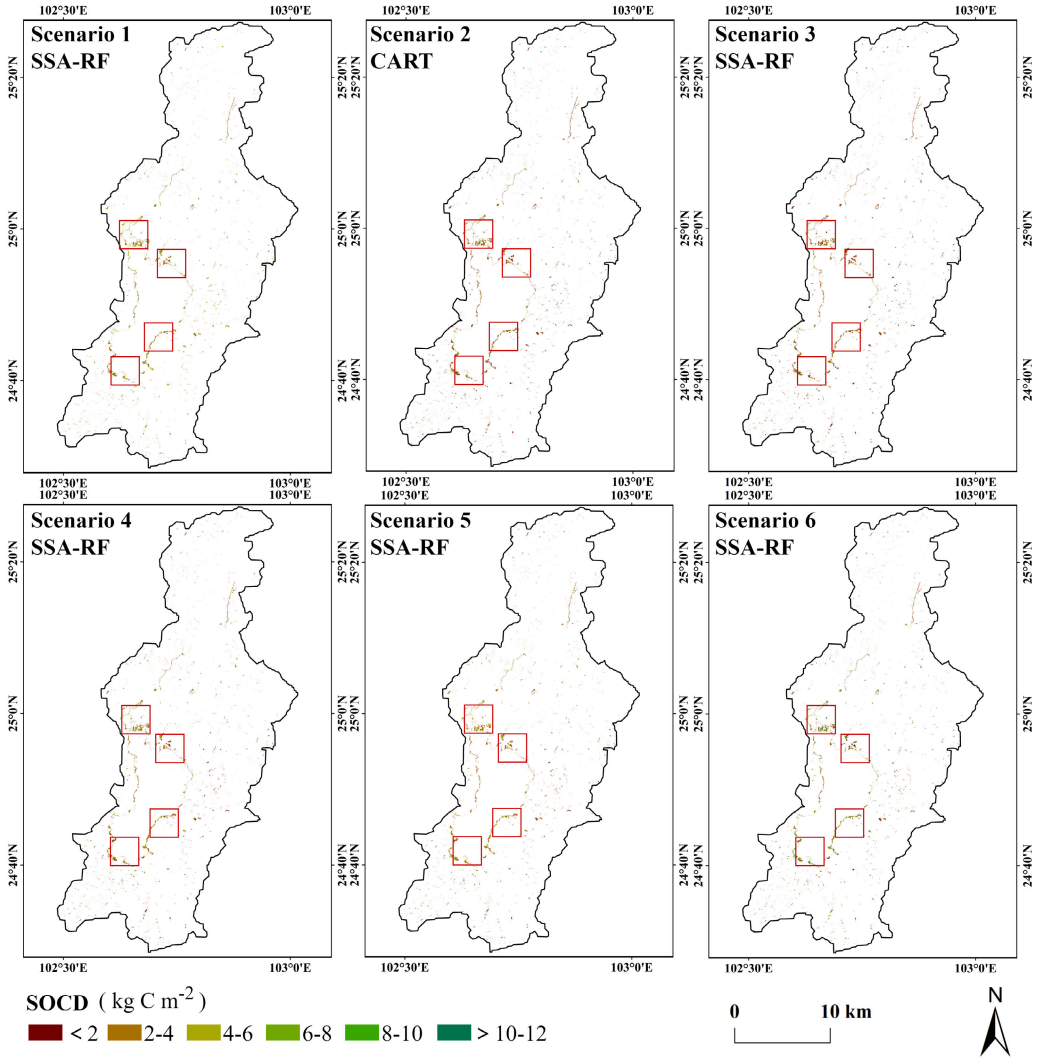


Fig. 8. SOCD spatial distribution-based different scenarios.

Specifically, the hyperspectral characteristics of S2 exhibit superior predictive performance over S1 in various data combinations. In these combinations, S2's hyperspectral features led to higher performance, with Scenario 2 achieving an R^2 of 0.60, compared to 0.47 for Scenario 1. Topographic and climatic features positively impact SOCD prediction. The inclusion of these features generally enhances the predictive performance of SSA-RF, SVM, and CART models. For instance, adding topographic features increased the SSA-RF model's performance by an R^2 of 0.03, aligning with findings by Ren et al. [45], which confirmed the effectiveness of terrain indicators in SOCD prediction. The consideration of both topographic and climatic features further improved the SSA-RF model's predictive value by 0.15, highlighting the significant role of climatic features in enhancing SOCD prediction. Future research should consider climatic features as a crucial factor for deeper quantitative analysis. Overall, the SSA-RF model outperforms the CART and SVM models in predicting SOCD in plateau swamp wetlands. In test scenario combinations, the SSA-RF model consistently exhibited the best performance as the number of features increased. The S1 + S2 + topographic + climatic SSA-RF model plays a positive role in predicting SOCD in swamp wetlands with small sample datasets. Future applications could extend this approach to other types of wetlands, such as rivers and lakes, where obtaining field samples is also challenging.

B. Soil Spectral Characteristics and Variable Importance

Our analysis of the reflectance characteristics from soil spectral measurements revealed that the correlation between the original spectral reflectance and SOCD was not significant. However, after mathematical transformations, the correlation between transformed soil spectral features and SOCD significantly improved, particularly following the reciprocal transformation, where the correlation peaked at 0.65. The strongest correlations were found within the 500–1400 wavelength range, corroborating findings by Song et al. [28]. Further analysis of variable importance in different models, both SSA-RF and CART, showed that features derived from S1 and S2 play a significant role in SOCD prediction. We also observed high importance for topographic and climatic features, consistent with our final prediction outcomes. Remote sensing technology provides

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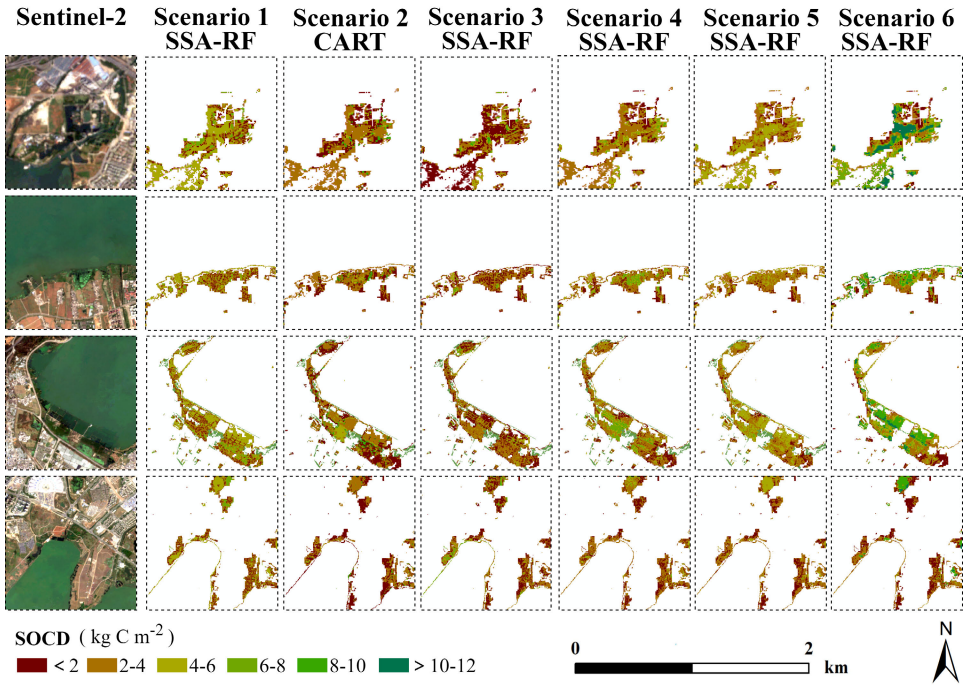


Fig. 9. Detailed views of SOCD distributions for different scenarios.

effective data support for predicting SOCD across different spatial scales. Previous studies, primarily focused on the remote sensing images themselves, often overlooked the importance of environmental variables. Our results underscore the positive role of topographic and climatic features in predicting SOCD. Specifically, slope affects the movement and accumulation of moisture and materials in the terrain, thus influencing the spatial distribution of SOCD [46]. Terrain not only controls the flow of solutes, water, and sediments, but also indirectly affects the formation of soil and the spatial distribution of its properties [47]. Additionally, precipitation and temperature significantly impact the distribution of plant communities and the accumulation process of soil organic matter, further affecting the distribution of SOCD [48], [49]. These findings provide a scientific basis for selecting predictive variables for SOCD.

C. Uncertainty

This study has demonstrated that in the complex terrain of plateau regions, the predictive ability of SOCD using multisource remote sensing data is superior to that of laboratory-measured soil spectral predictions. However, there are still uncertainties worth considering. On the one hand, despite strict control of laboratory conditions and elimination of external light sources, the complexity of the laboratory spectral collection process, such as the frequency of instrument calibration and changes in ambient temperature, may introduce uncertainties in this article. On the other hand, given that the research area is located in a high plateau mountainous region with extensive cloud cover throughout the year, there are temporal discrepancies between soil sampling and satellite data acquisition that also introduce uncertainties. In the future, we will strictly control the timing of soil sampling and satellite data collection to minimize the impact of these discrepancies.

VI. Conclusion

This study proposed a framework to predict SOCD in plateau swamp wetlands by combining multisource remote sensing data with the SSA-RF algorithm (S1 + S2 + topographic + climatic SSA-RF). Applied to the Dianchi Basin in the Yunnan–Guizhou Plateau, this model demonstrated superior predictive accuracy, achieving an R^2 of 0.76, RMSE of 1.14, MAE of 0.65, and RPD of 1.98. It significantly outperformed the laboratory spectral model, with improvements of 0.24 in R^2 and 0.5 in RPD. Moreover, it exceeded the performance of the S1 + S2 model, with increases of 0.15 in R^2 and 0.39 in RPD, and the S1 + S2 + topographic model, with gains of 0.12 in R^2 and 0.38 in RPD. Employing the S1 + S2 + topographic + climatic SSA-RF model, the TOC stock of the soil in the Dianchi Basin swamp wetlands was estimated at 1.32×10^5 t. This article provides a reference framework for measuring SOC in fragmented plateau swamp wetlands, significantly contributing to studies on wetland carbon stock.

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